

STAA57-Winter 2024: Worksheet-7

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Questions 1:

Shoveling is not exactly a high-tech activity, but will continue to be a required task even in our information age. The article “A Shovel with a Perforated Blade Reduces Energy Expenditure Required for Digging Wet Clay” (*Hum. Factors*, 2010: 492–502) reported on an experiment in which each of 13 workers was provided with both a conventional shovel and a shovel whose blade was perforated with small holes. The authors of the cited article provided the following data on stable energy expenditure [kcal/kg(subject)/lb(clay)]:

<i>Worker:</i>	1	2	3	4	5	6	7
<i>Conventional:</i>	.0011	.0014	.0018	.0022	.0010	.0016	.0028
<i>Perforated:</i>	.0011	.0010	.0019	.0013	.0011	.0017	.0024
<i>Worker:</i>	8	9	10	11	12	13	
<i>Conventional:</i>	.0020	.0015	.0014	.0023	.0017	.0020	
<i>Perforated:</i>	.0020	.0013	.0013	.0017	.0015	.0013	

Test the hypothesis that the two shovels perform equally, with the title of the article as the alternative.

Make sure to write your hypothesis, and explain briefly what test you will be performing and why.

Questions 2:

The accompanying data on response time appeared in the article “The Extinguishment of Fires Using Low-Flow Water Hose Streams—Part II” (*Fire Techn.*, 1991: 291–320). The samples are independent, not paired.

<i>Good</i>	.43	1.17	.37	.47	.68	.58	.50	2.75
<i>visibility</i>								
<i>Poor</i>	1.47	.80	1.58	1.53	4.33	4.23	3.25	3.22
<i>visibility</i>								

Using a permutation test, test the hypothesis that the average response is same for the two types of visibility. Use not equal as an alternative.

Instructions for completing this worksheet:

- Create a R markdown file, use your name and student id as the author name.
- Use the command ‘\pagebreak’ between your answers to Q1 and Q2 (this will make sure when you convert to pdf, answer to Q2 starts on a new page)
- Create an html output first, open it in any browser and then print or save as a pdf. (if you already know how to create a pdf directly from R studio then disregard this step)
- Upload your pdf to crowdmark in the place designated for Q1.
- Drag the pages corresponding to Q2 to the place designated to Q2.
- Finish submitting everything by 6pm please.